

Figure 8.2 Automated web tests have a somewhat justified reputation of being slow (courtesy of http://xkcd.com).

reasons, it's virtually impossible to do good BDD-style acceptance testing with Record-Replay tools.

Fortunately, web browser automation using open source tools like Selenium WebDriver has improved with age, and today it's very possible to write reliable, robust, and maintainable automated web tests, particularly if you apply the principles of layers and reusable steps that we'll discuss in this chapter. Isolating the code that interacts with the web page in a single class or method (using the Page Objects pattern, for example) goes a long way toward making these tests easier to maintain.

But no matter what tooling you use, the problem of speed still remains (see figure 8.2). An automated web test needs to open a browser and reproduce the actions of the user through the browser, which takes time. Each page load slows down the test. For modern web applications using AJAX-based libraries like AngularJS and Backbone.js, the page updates tend to be much faster, so speed is less of an issue. But, in general, automated web testing will always be significantly slower than tests that exercise the application code directly.

8.1.2 Web testing with headless browsers

One strategy for addressing the issue of slow web tests is to use a headless browser. Headless browser libraries such as HtmlUnit for Java (http://htmlunit.sourceforge.net), Webrat for Ruby (https://github.com/brynary/webrat), and Twill for Python (http://twill.idyll.org) send HTTP queries directly to the server, without having to start up an actual web browser like Firefox or Chrome. HtmlUnit, for example, works with WebDriver and a number of other Java-based web-testing libraries, providing APIs to analyze the structure of the HTML document. PhantomJS (http://phantomjs.org) provides a more accurate browser simulation, because it renders the HTML like a real browser would, but does so internally. Headless tests often run more quickly than they would using a real browser, and they also make it easier to run a number of tests in parallel.

Some of these libraries, such as HtmlUnit, have limited support for AJAX and JavaScript, so your mileage with a modern JavaScript-based website may vary. This can